

Code-Layout, Readability and Reusability

Date	20 june 2026
Team ID	SWTID-2026-9290
Project Name	A COMPHRENSHIVE MEASURE OF WELL BEING
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	spacing/tabs used throughout the code	yes	Enhances structural clarity and immediate readability.
2	Proper File Structure	Files and folders are logically organized	yes	Project files are organized logically into appropriate folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Variables such as wellBeingScore, mentalHealth, physicalHealth clearly describe their purpose
4	Function / Method Names	Functions are descriptively named	yes	Functions like calculateWellBeing(), displayResult(), and getUserInput() are descriptive.
5	Code Comments	Inline and block comments explain logic	yes	Comments explain important logic and improve code understanding
6	Modular Design	Code is split into reusable functions/modules	yes	Code is divided into reusable functions/modules for better maintainability.
7	No Redundant Code	Duplicate or unused code is removed	yes	Duplicate and unnecessary code has been removed
8	Error Handling	Exceptions and errors are handled gracefully	yes	Input validation and exception handling ensure smooth execution.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Health check end point	Node.js / Express	All microservices	high

2	authmiddleware	Python / FastAPI	User and admin gateways	high
3	loggrservice	Python / FastAPI	Payment and notification engines	high
4	Well-being score calculator	Phython/fastAPI	Physical, Mental, Social, Financial, and Emotional Well-Being modules	high
5	Input validation and error handling's	Phython/fast API	All API endpoints and forms	medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean indentation, logical module organization, and consistent naming conventions throughout the codebase.
Readability	4	Descriptive variable names make the logic transparent, though deeply nested loops could be flattened.
Reusability	4	Functions are modular and follow single-responsibility principles, though hardcoded config paths limit portability.
Documentation / Comments	5	Docstrings clearly define inputs, outputs, and edge cases without cluttering the actual execution logic.
Overall Score	4.5	Docstrings clearly define inputs, outputs, and edge cases without cluttering the actual execution logic.